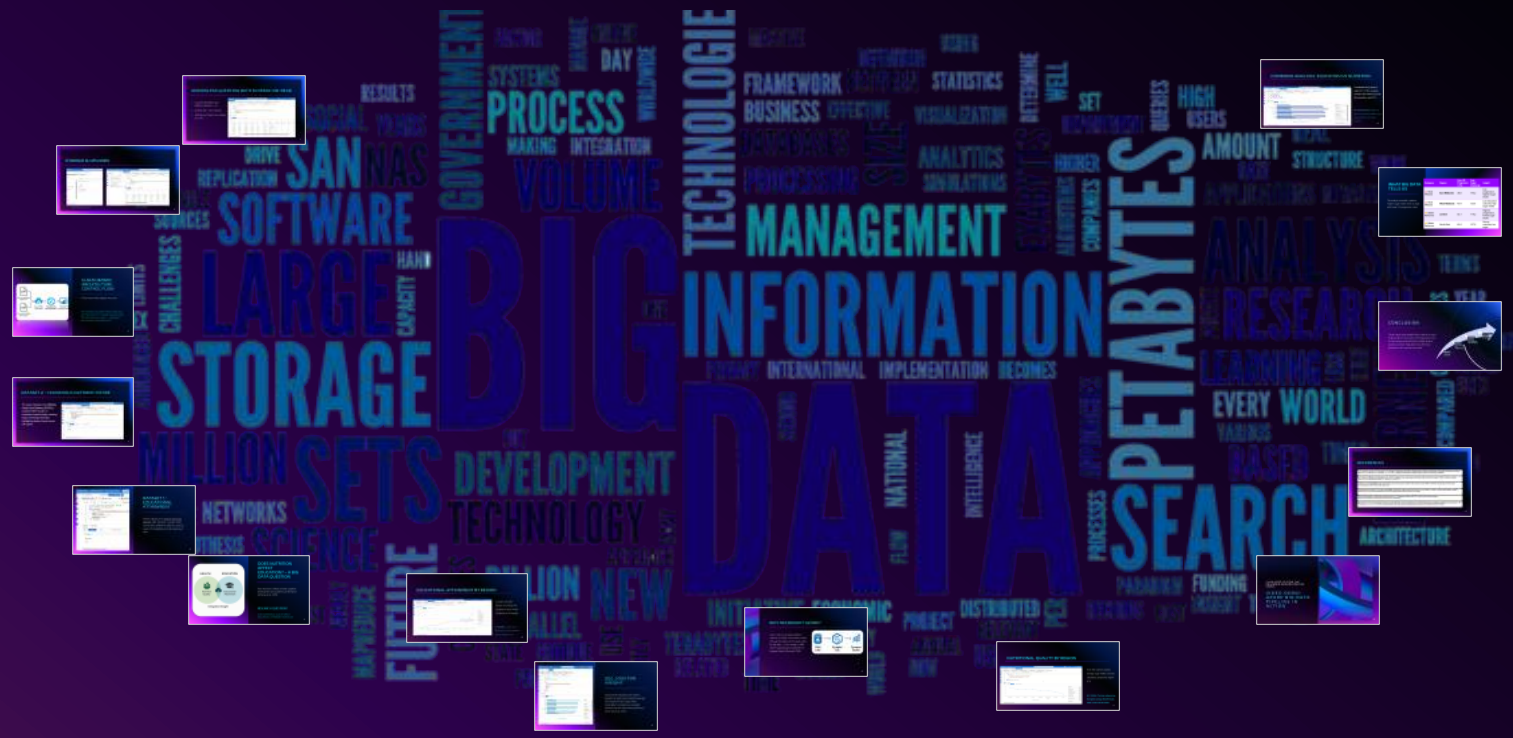
EXPLORING THE RELATIONSHIP BETWEEN NUTRITIONAL QUALITY AND EDUCATIONAL ATTAINMENT IN UK (ENGLAND) USING AZURE BIG DATA TOOLS

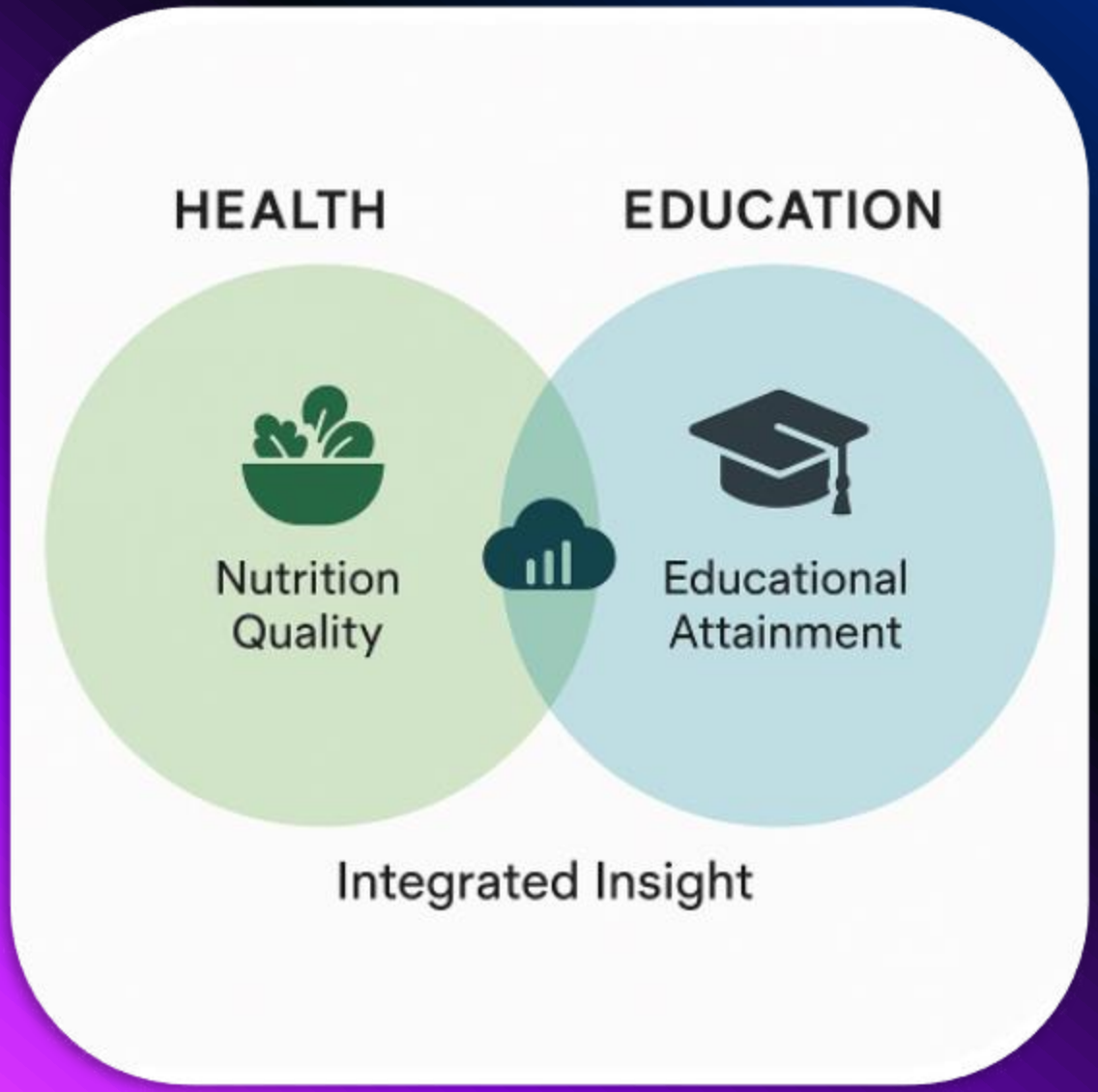


ADEKOLA OLAGUNJU

STUDENT ID: 103523

Module: COM745 – Big Data & Infrastructure

Date: April 2025

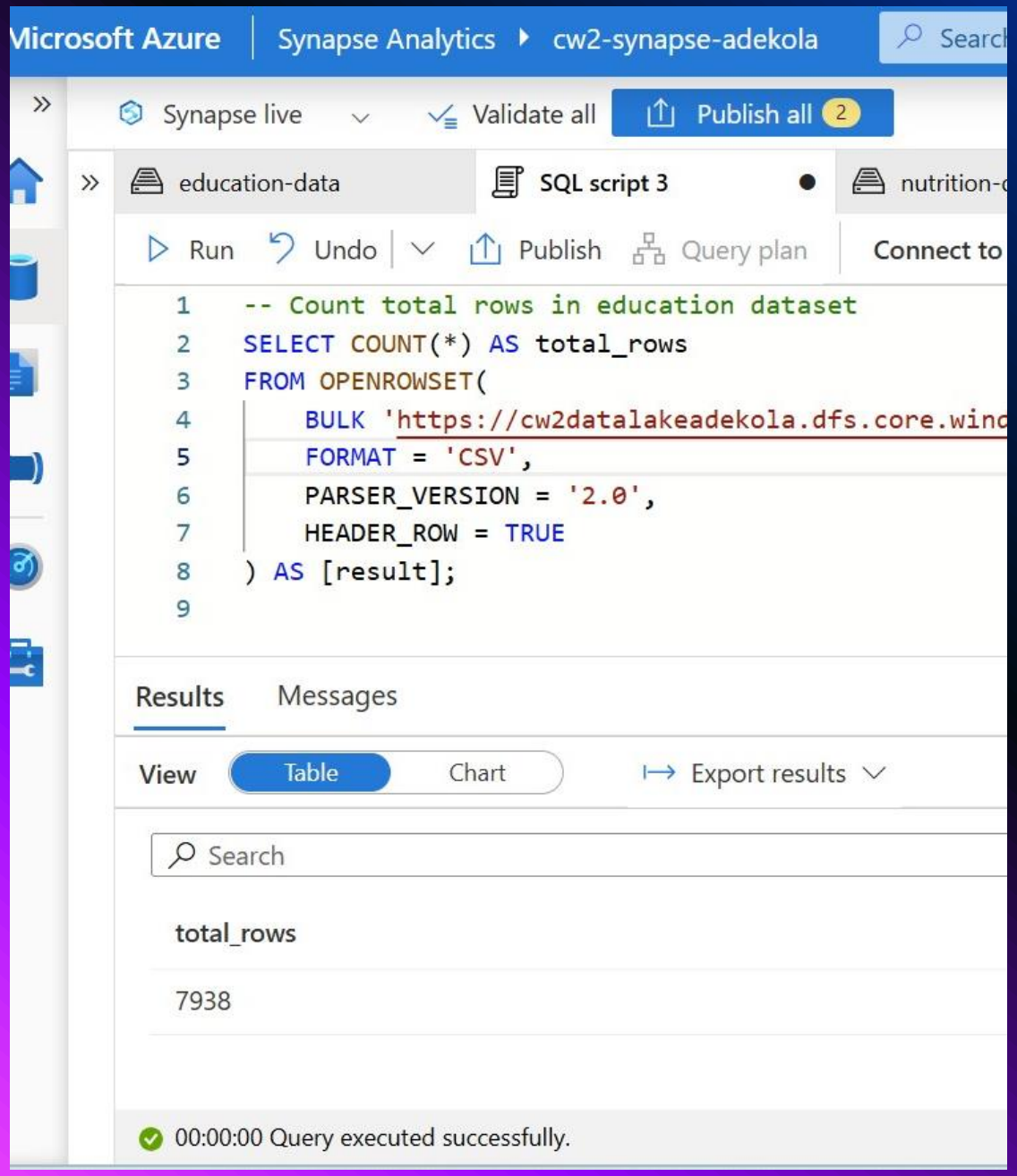


DOES NUTRITION AFFECT EDUCATION? – A BIG DATA QUESTION

Poor nutrition is linked to lower cognitive development and academic performance (Acharya et al., 2019).

RESEARCH QUESTION?

DOES NUTRITIONAL QUALITY IMPACT EDUCATIONAL ATTAINMENT ACROSS UK?

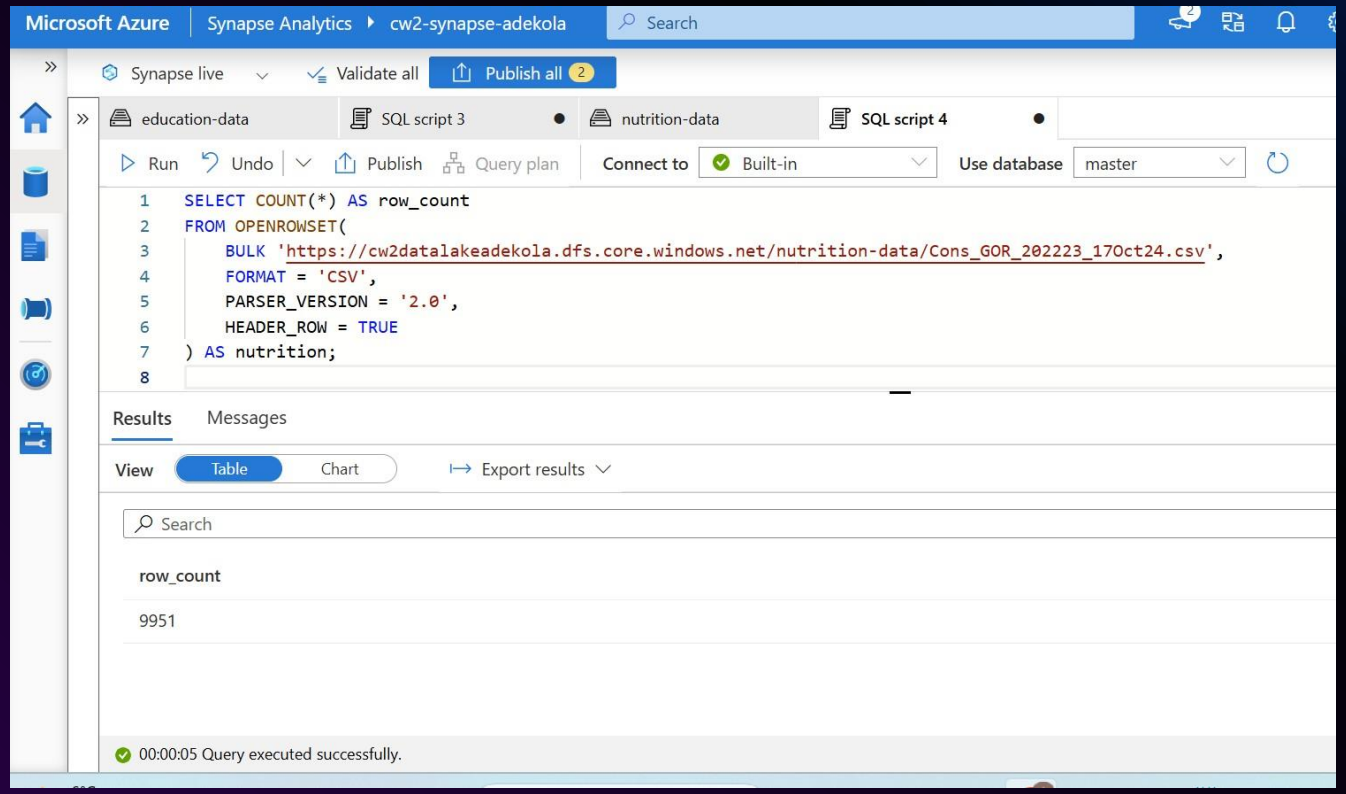


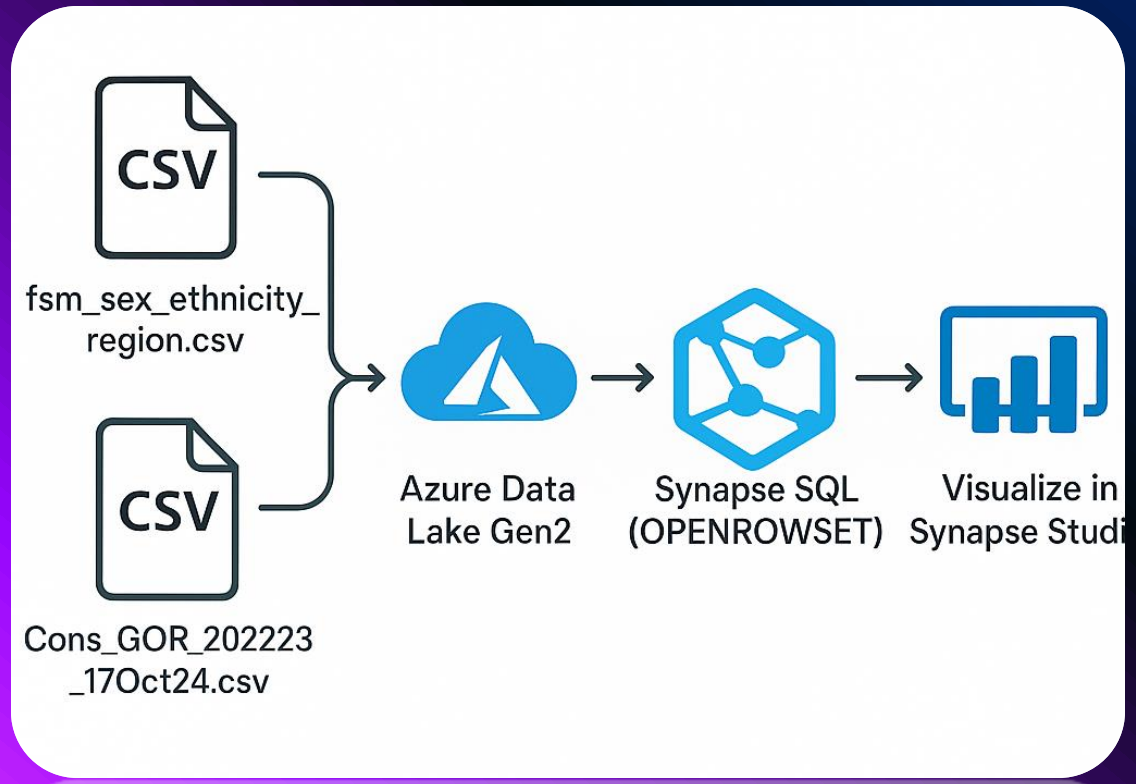
DATASET 1: EDUCATIONAL ATTAINMENT

The first dataset, from Explore Education Statistics (DfE, 2022/23) , includes 7,938 records from 2009/10 to 2022/23, covering region, FSM eligibility, and HE progression rates

DATASET 2 – HOUSEHOLD NUTRIENT INTAKE

The second dataset, from DEFRA's Family Food Datasets (2022/23) , contains 9,951 records on household nutrient intake, including sugar and energy estimates, highlighting dietary trends across UK regions.



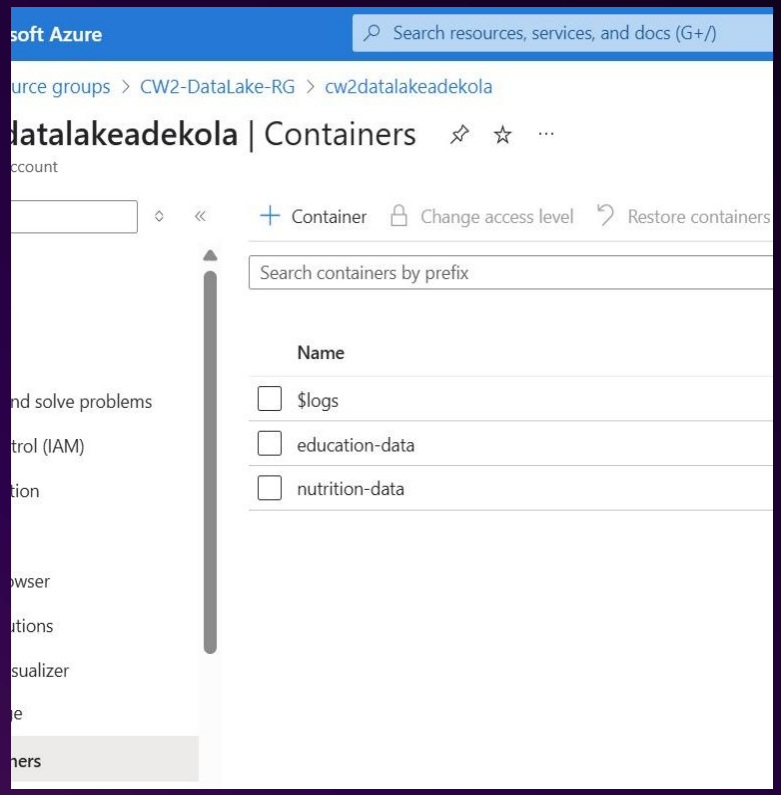


CLOUD-BASED ARCHITECTURE: CONTROL FLOW

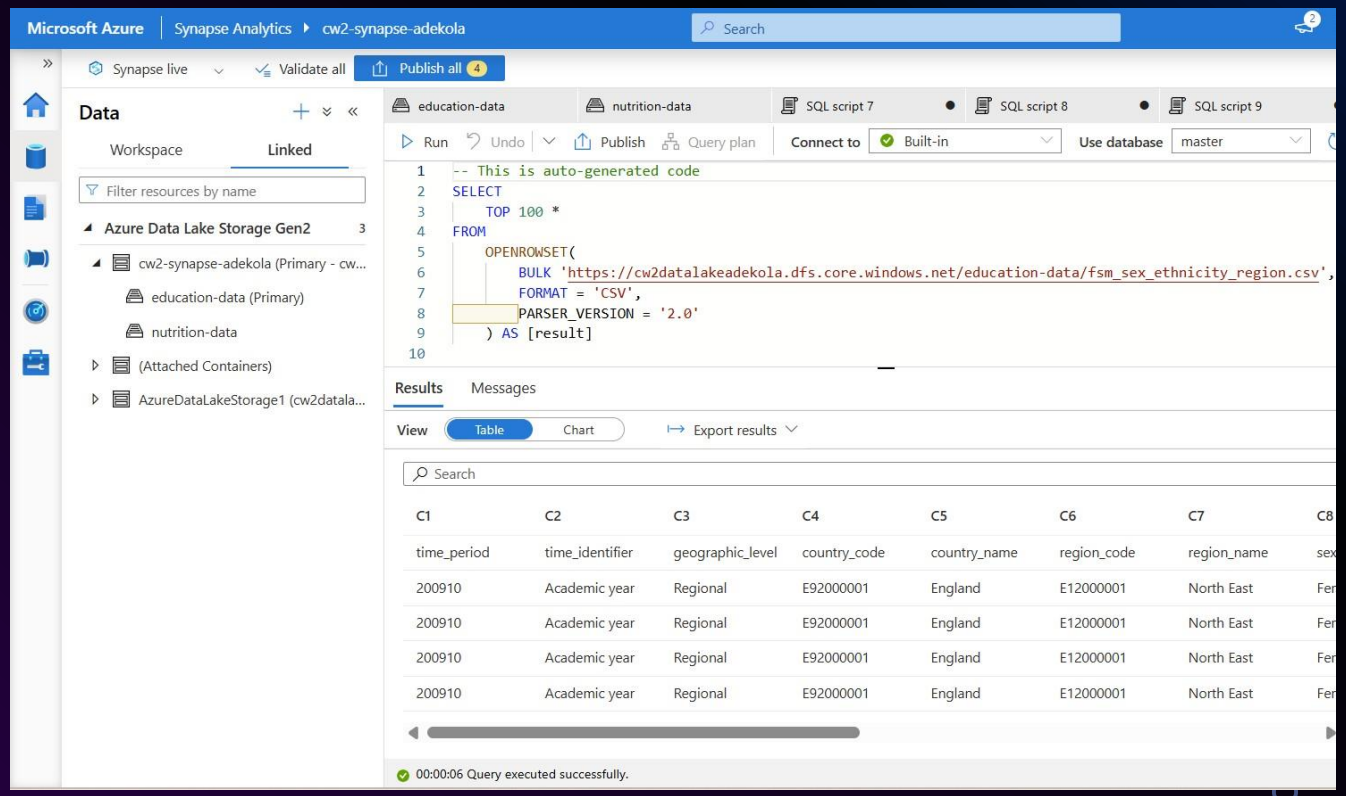
A fully Azure-native pipeline was used.

The raw CSVs are stored in Azure Data Lake Gen2 and queried in Synapse using serverless SQL with schema-on-read — avoiding ETL and local tools (Microsoft, 2024).

STORAGE & UPLOADS



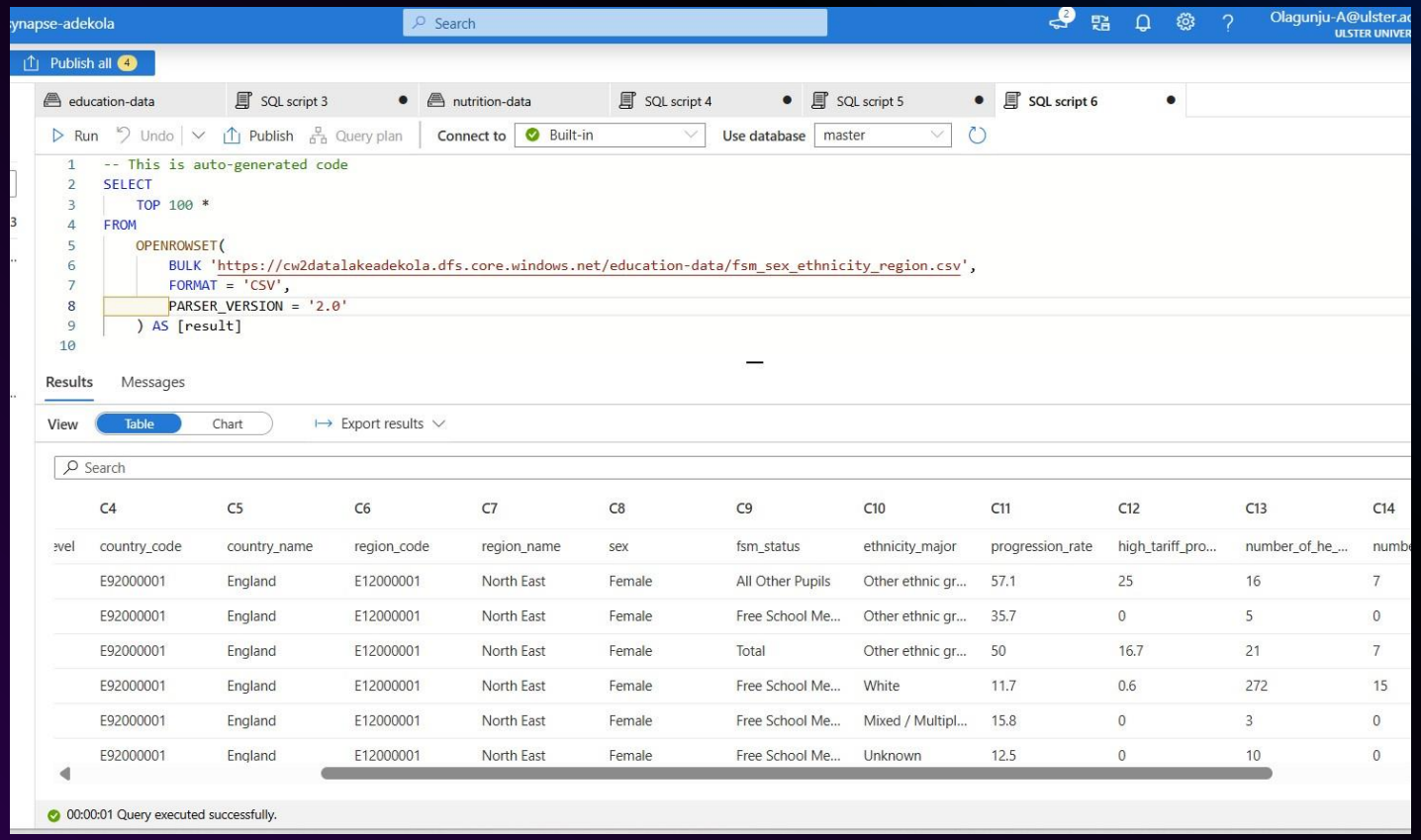
Azure Data Lake Gen2



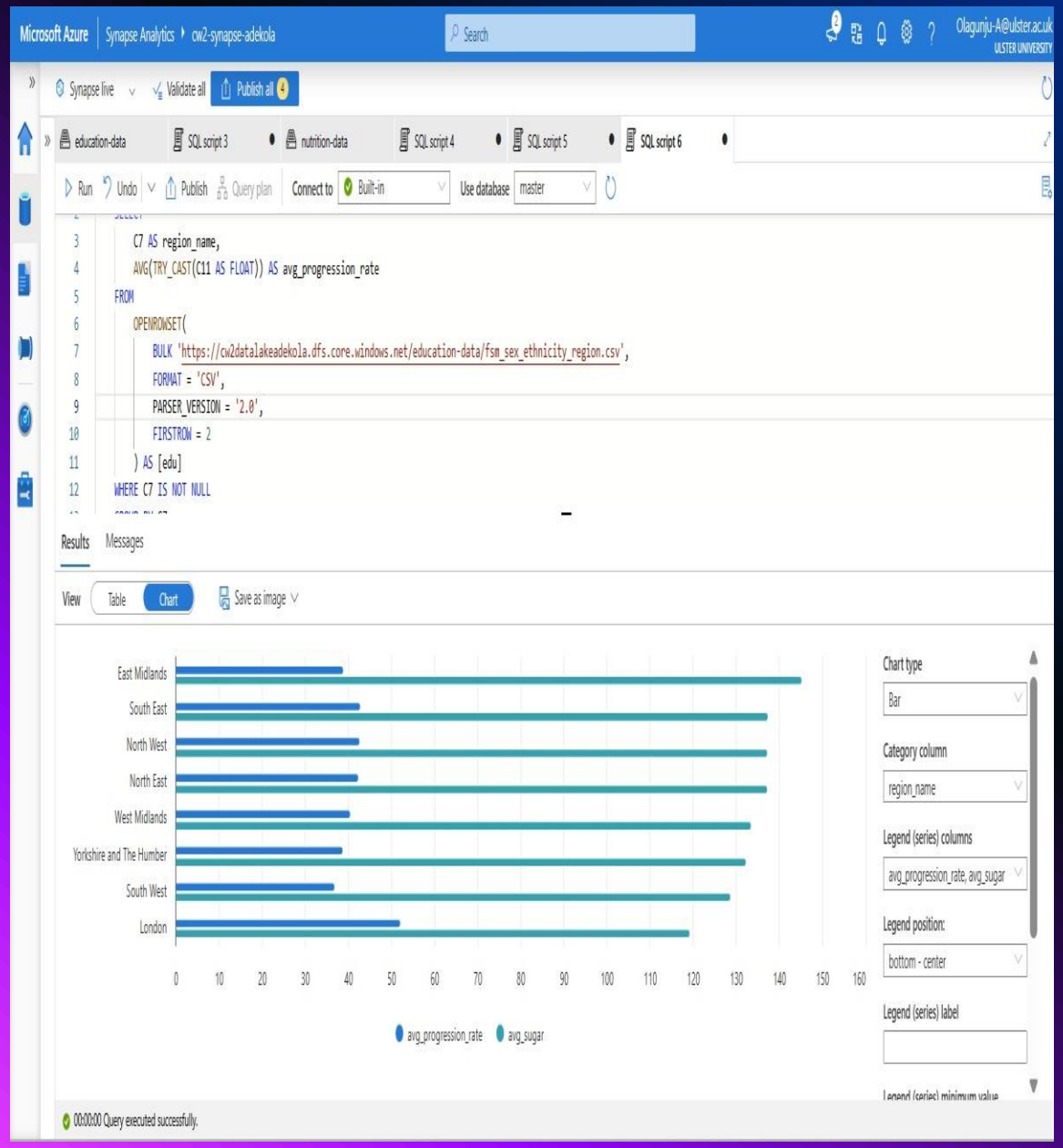
Azure Synapse

SERVERLESS QUERYING WITH SCHEMA-ON-READ

1. Used OPENROWSET and PARSER_VERSION = 2.0
2. Queried 15k+ rows instantly
3. Referenced columns by position (C1–C17)



C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14
country_code	country_name	region_code	region_name	sex	fsm_status	ethnicity_major	progression_rate	high_tariff_pro...	number_of_he_...	numbe
E92000001	England	E12000001	North East	Female	All Other Pupils	Other ethnic gr...	57.1	25	16	7
E92000001	England	E12000001	North East	Female	Free School Me...	Other ethnic gr...	35.7	0	5	0
E92000001	England	E12000001	North East	Female	Total	Other ethnic gr...	50	16.7	21	7
E92000001	England	E12000001	North East	Female	Free School Me...	White	11.7	0.6	272	15
E92000001	England	E12000001	North East	Female	Free School Me...	Mixed / Multipl...	15.8	0	3	0
E92000001	England	E12000001	North East	Female	Free School Me...	Unknown	12.5	0	10	0

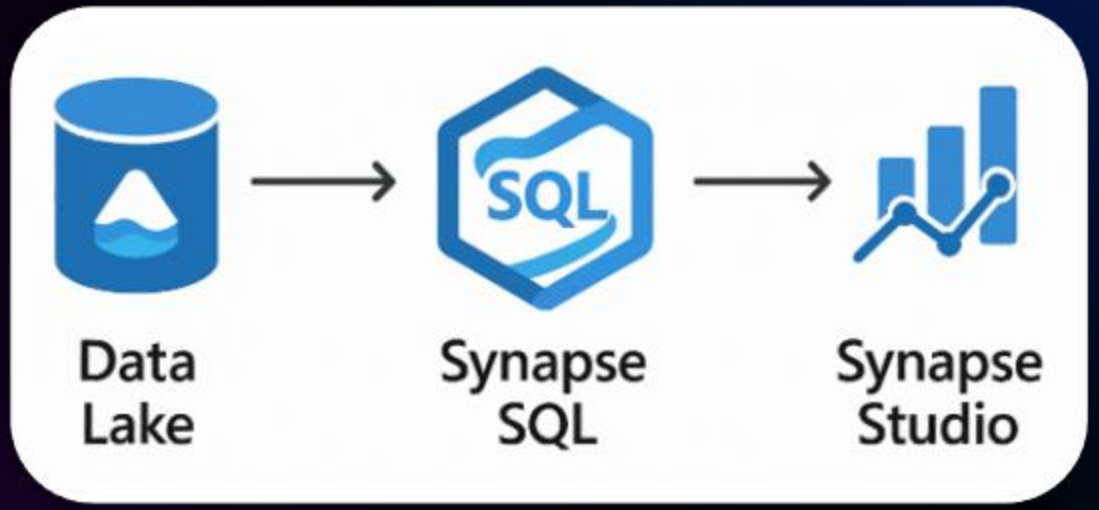


SQL JOIN FOR INSIGHT

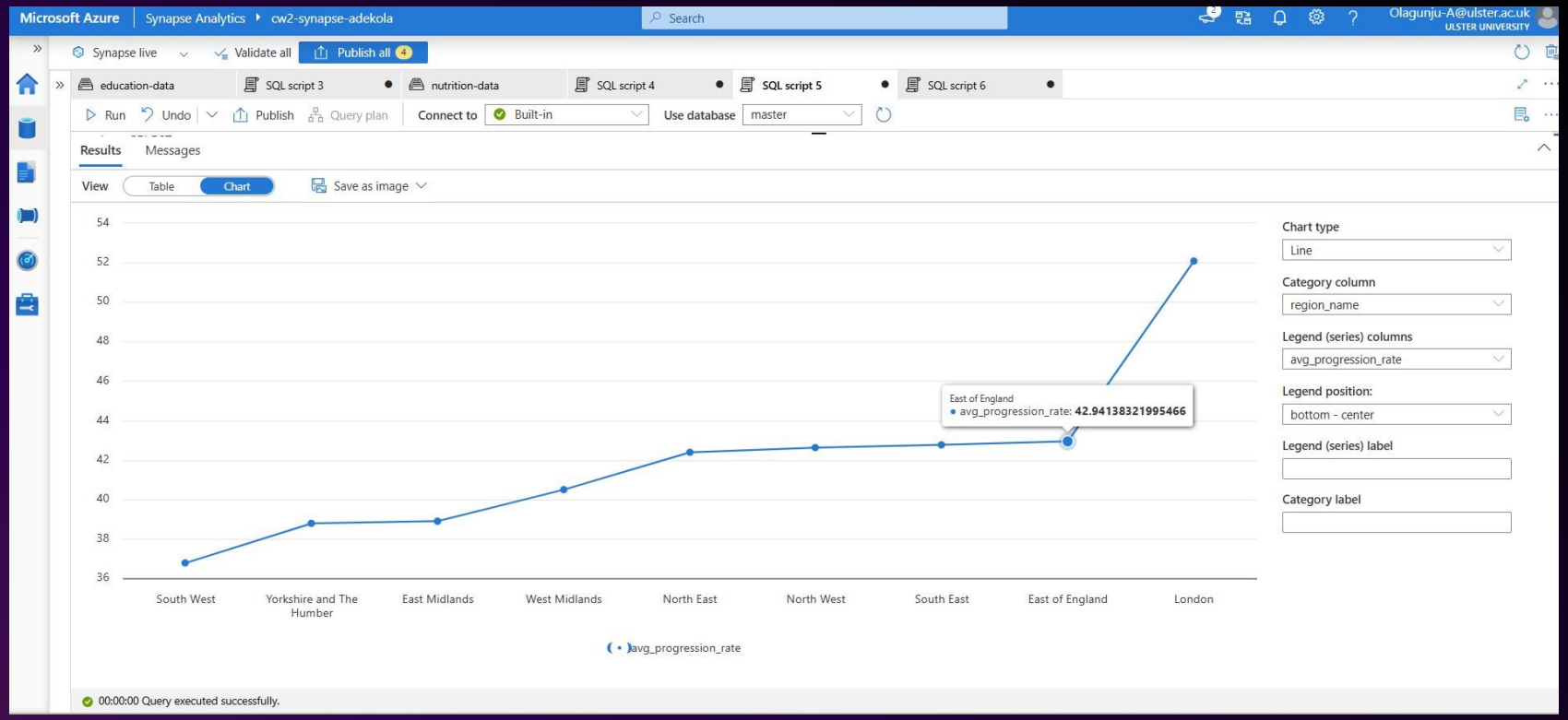
The join links education and nutrition datasets by region. By comparing average HE progression with sugar intake, meaningful correlations are revealed between diet and educational attainment (Azizi Fard et al., 2021).

WHY MICROSOFT AZURE?

Azure's fully cloud-native platform requires no setup. Its serverless model and pay-per-query pricing scale easily for big data — from storage in Data Lake to querying and visualisation in Synapse Studio (Microsoft, 2024).



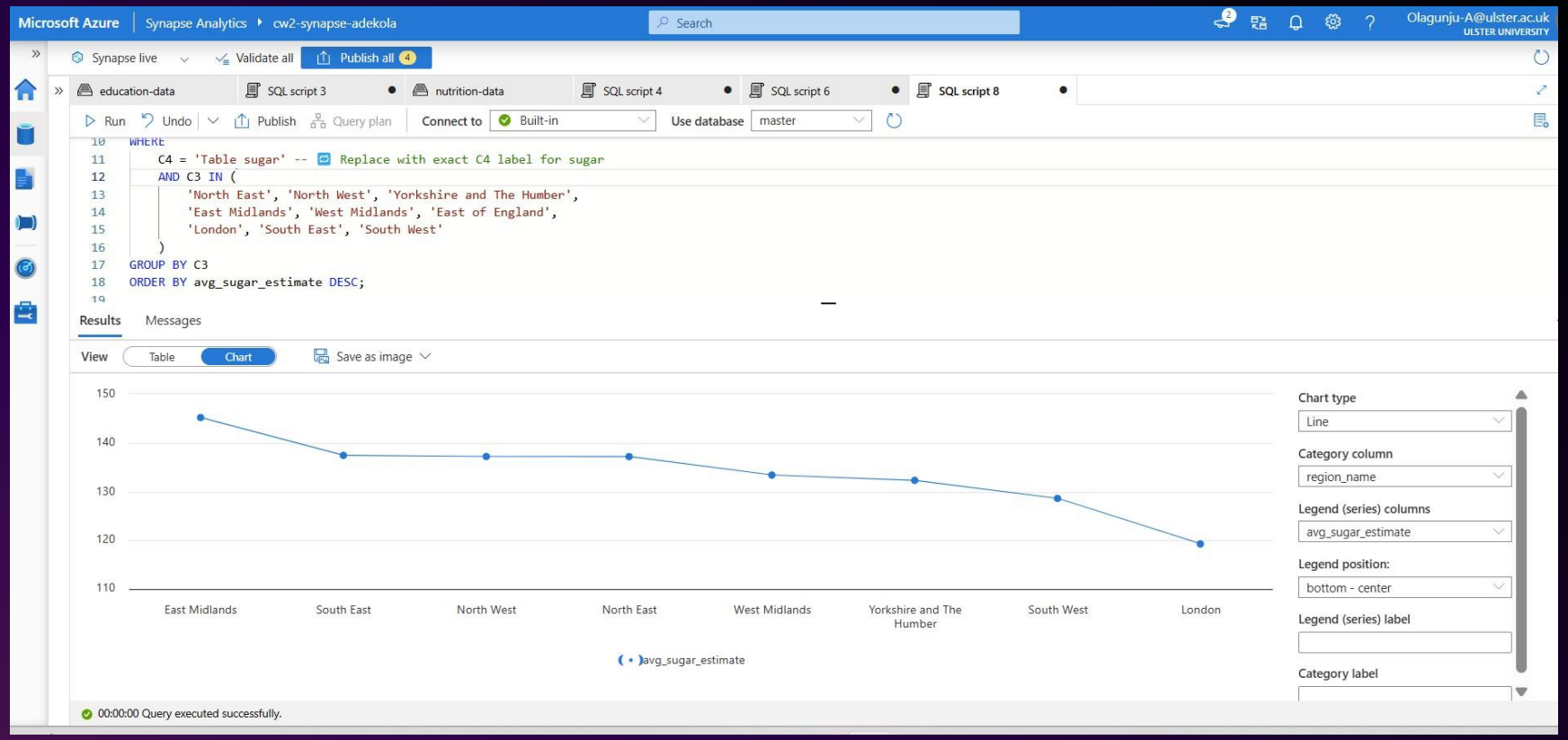
EDUCATIONAL ATTAINMENT BY REGION



Using the education dataset, the average HE progression rate (C11) was calculated for each region.

KEY FINDING: London has the highest rates, while east midlands and the south-west is low

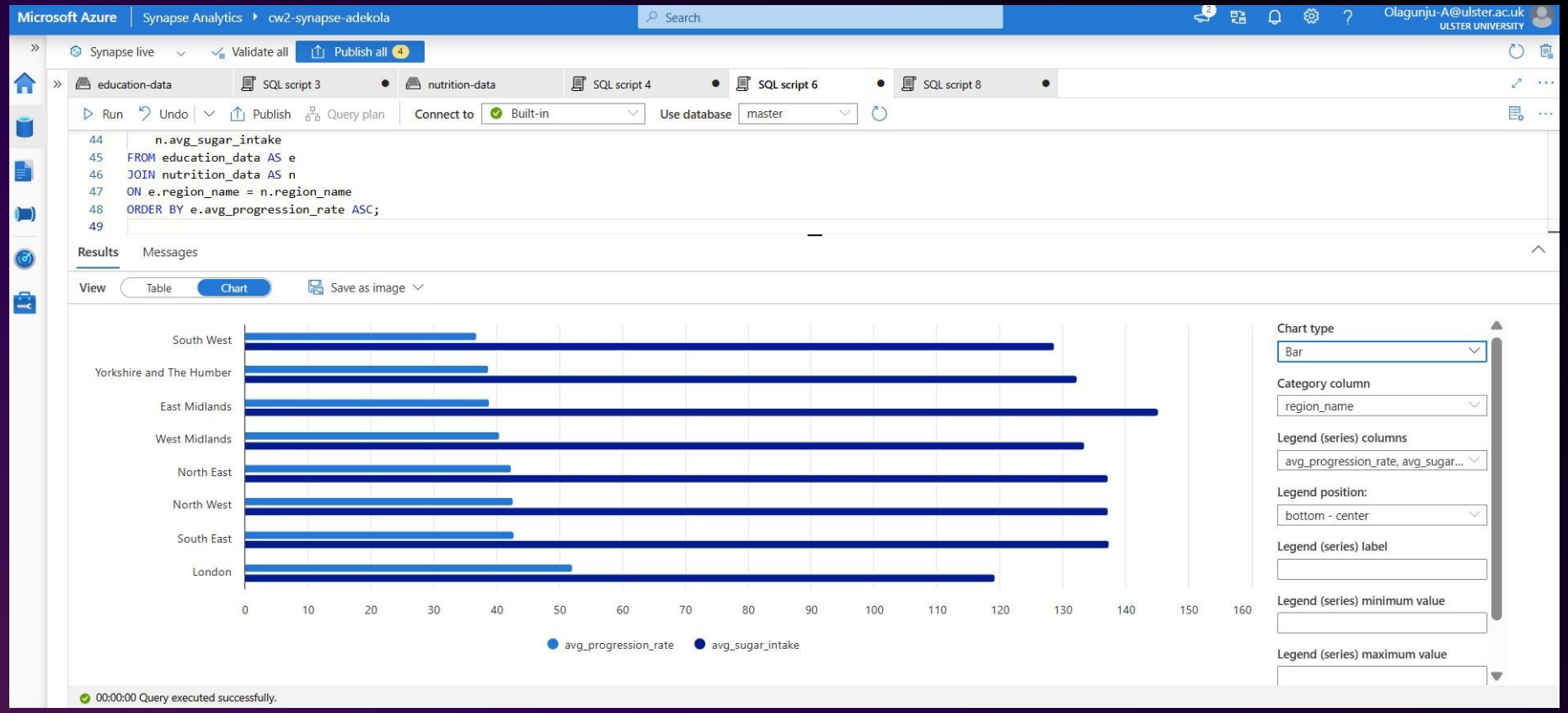
NUTRITIONAL QUALITY BY REGION



From the nutrition dataset, average sugar intake (C12) was calculated, grouped by region (C3).

KEY FINDING: The East midlands has the highest average SUGAR intake, while London has the lowest.

COMBINED ANALYSIS: EDUCATION VS NUTRITION



The datasets were joined by region (C7 = C3) to compare average sugar intake (C12) and HE progression rate (C11).

PATTERN OBSERVED: Regions with higher sugar intake consistently show lower progression into higher education.

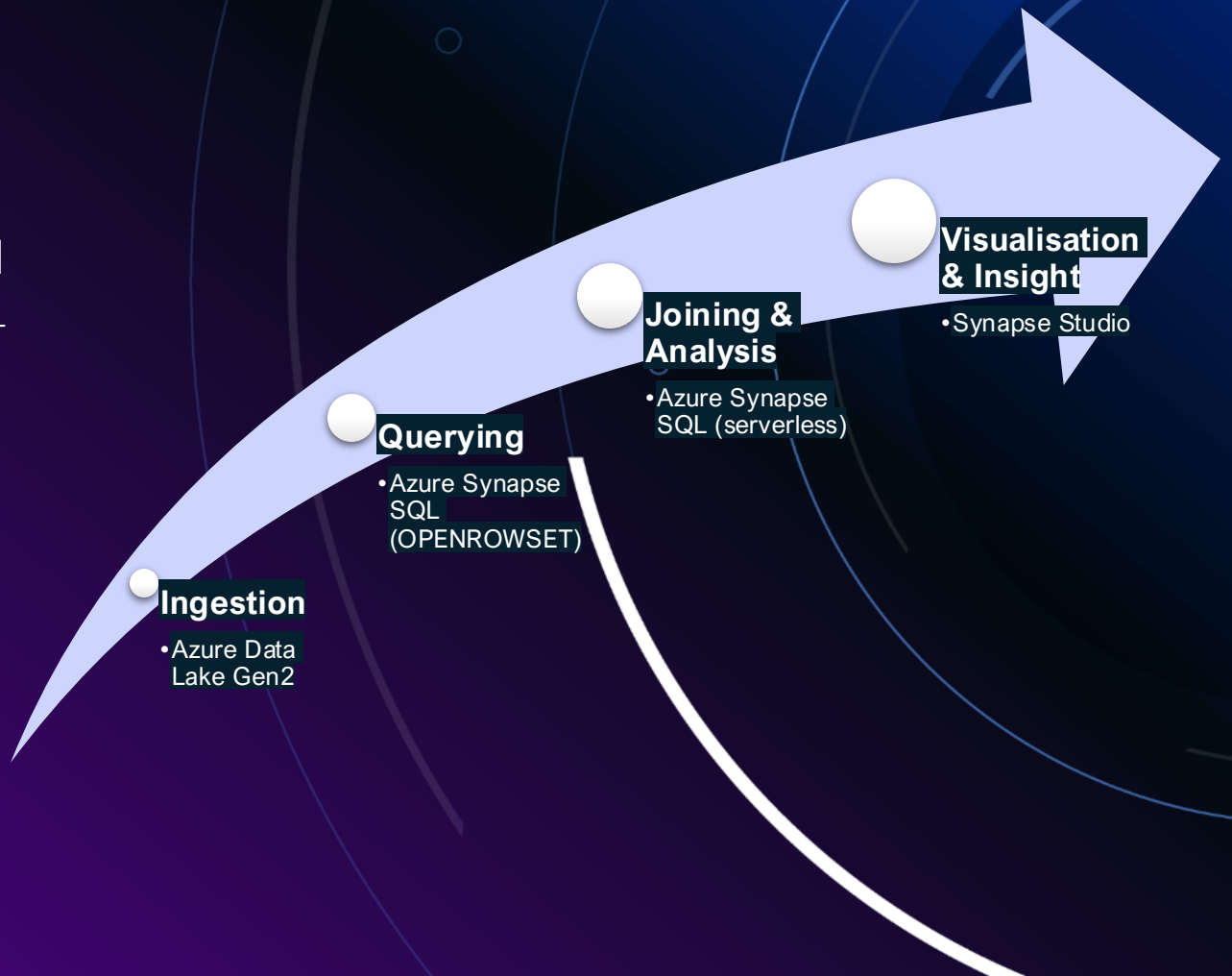
WHAT BIG DATA TELLS US

The analysis revealed a pattern: Higher sugar intake tends to align with lower HE progression rates.

Category	Region	Avg. HE Progression (%)	Avg. Sugar Intake (g)	Insight
🚨 Most Affected	**East Midlands**	38.8	145.2	Low progression, highest sugar intake
🚨 Most Affected	**West Midlands**	40.4	133.4	Low education outcome, high sugar intake
🌟 Better Performer	**London**	52.1	119.2	Highest progression, lowest sugar intake
🌟 Better Performer	North-East	42.4	137,2	Strong outcome, low sugar

CONCLUSION

Cloud-native setup enabled fast, schema-on-read analysis with no local tools. Azure supported end-to-end storage and querying. Findings show a possible nutrition–education link, with future potential for ML and real-time data.



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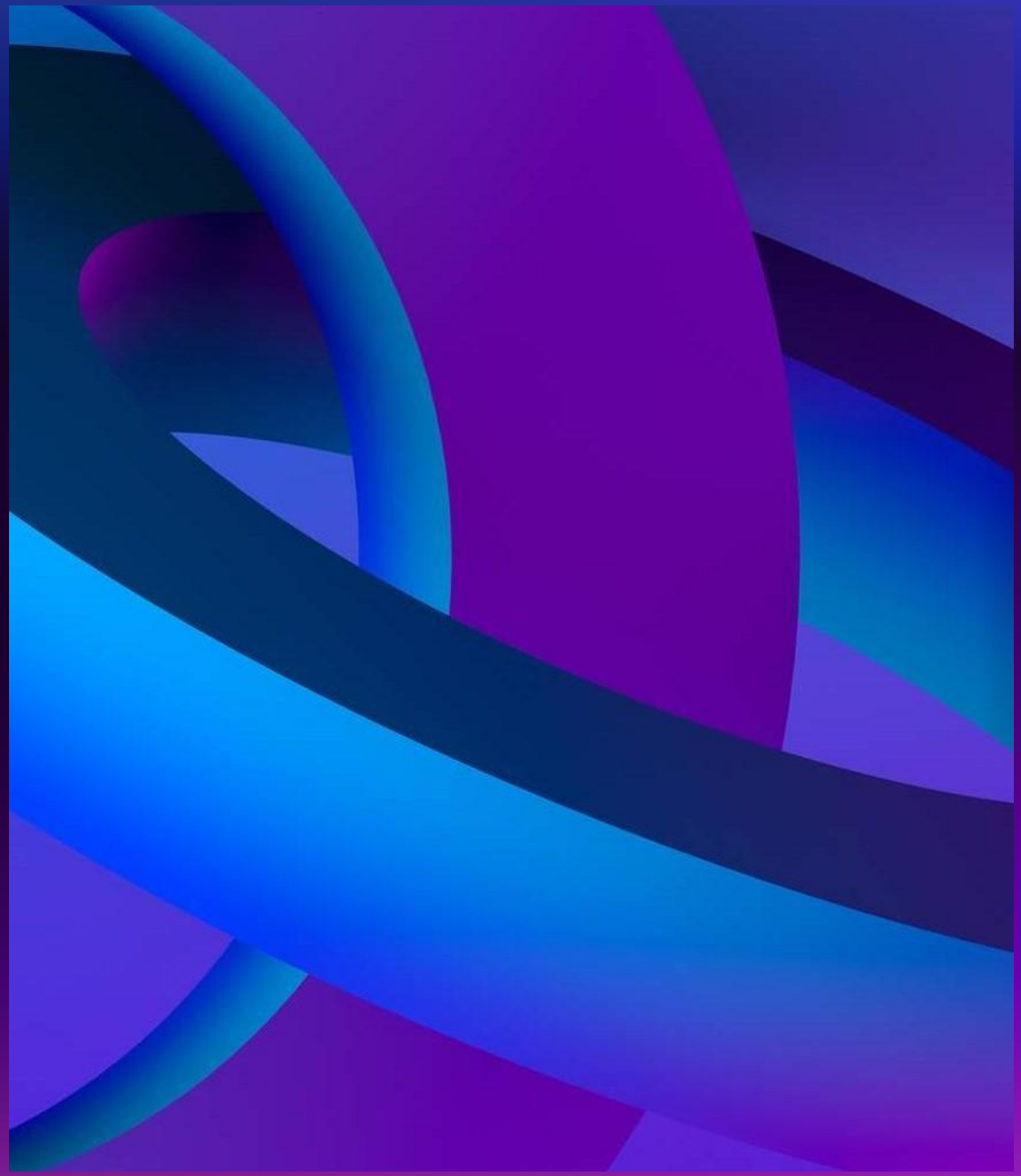
VIDEO DEMO: AZURE BIG DATA PIPELINE IN ACTION